Western Digital: qualified capacity and the value of stored AI data

Western Digital supplies a scarce physical resource: reliable storage capacity that large customers have already qualified. Growing AI activity leaves more data to retain, while new drive platforms take time to manufacture and validate. Higher net selling prices can therefore increase earnings faster than sales, provided customers accept the drives and the premium reaches the manufacturer.

The investment edge is the connection between qualified exabytes, contract renewals and profit after real costs. The October 20 broker checks suggest greater pricing tension. Most of the shortage thesis was already represented, so the revised case replaces the price endpoint inside that existing finite branch and keeps transmission risk visible.

Current analytical output	Value	Meaning
Q4 2027 model ceiling	USD 819 per share	Q4 2027 through Q3 2028 adjusted operating earnings
Q1 2028 model ceiling	USD 873 per share	Q1 through Q4 2028 adjusted operating earnings
Valuation method	20x / 388 million diluted shares	Starting-quarter denominator; no assumed buyback benefit
Analytical score	46 / 60; 77 / 100	Seven categories reassessed; distinct from Excel control
Portfolio Excel Base	77 / 100	Protected existing owner control

Research date: October 21, 2026. The pricing report is dated October 20 at 03:44 UTC, or October 19 late evening in Toronto. Latest completed issuer results were released August 5 for the quarter ended July 3. The October 22 earnings event is still future. [HDD-P2; WDC-1; WDC-9]

Financial tables use USD millions and whole-number display unless stated. Nonzero amounts below USD 1 million appear as <1; true zeros remain 0. Shares are millions; physical quantities are exabytes (EB); prices are USD per terabyte (TB), with small price changes shown in whole US cents per TB. Nonzero rates below 1% appear as <1%. Rounding can prevent displayed subtotals from adding exactly. The model and precision appendix retain the exact calculations. The ceiling is an analytical estimate, with substantial execution and valuation uncertainty.

Historical header snapshot: USD 448 per share at the September 21, 2026 close, rounded for reading; approximately 361 million basic shares at August 7. These dated observations do not enter the forward earnings denominator or score. Exact source snapshot is retained in the precision supplement. [WDC-2; WDC-6]

Business perimeter, bottleneck and competition

Western Digital has operated as a standalone hard-disk-drive company since the February 2025 SanDisk separation. Cloud, enterprise, client and consumer HDD sales form the continuing perimeter. Sandisk flash sales and the monetized retained stake supply no forward operating earnings here. [WDC-1; WDC-2]

The bottleneck combines heads and media engineering, manufacturing yield, firmware, reliability and customer qualification. More storage per platter increases shipped bytes without an equivalent increase in finished-drive units. WDC can advance its ePMR and UltraSMR platform while qualifying heat-assisted magnetic recording (HAMR). Each platform must clear customer acceptance and cost hurdles before revenue and profit are earned.

Seagate is a scaled competitor with commercial HAMR experience. Toshiba is the third major HDD supplier and has begun shipments from its expanded Philippine nearline line. SSDs compete in performance-sensitive storage and can gain applications if NAND prices fall enough. Those alternatives constrain pricing and prevent monopoly credit. A small new supplier cannot easily reproduce the full qualified head/media and reliability platform. [WDC-2; TOS-1]

Opportunity phase	Company capture	Boundary
2026-27: present cloud storage	40TB ePMR adoption and selective contract repricing	Existing storage wallet; no separate AI spending multiplier
2027-28: HAMR and higher density	Accepted 44TB shipments and later performance features	Newer products replace older drive capacity
2029-30: retained-data growth	Larger data lakes and wider enterprise/cloud deployment	Long-dated demand, yield and margins remain LOW confidence

Persistent inference context, video and physical-AI data can expand the need for retained storage. A WD-sponsored IDC survey found 61% of respondents had AI-related data growth of at least 25% over the prior year. This supports direction; the survey is neither an industry sales forecast nor a WDC order. [WDC-8]

WDC / OCTOBER 21, 2026 / FRAMEWORK 8.0

Latest actual quarter and upcoming guidance

The July 3 quarter is WDC fiscal Q4 2026 and is placed in the calendar Q2 bucket. Sales of USD 3,747 million reconcile to USD 3,335 million cloud, USD 225 million client and USD 187 million consumer. Management reported 231 total EB; the precise cloud-only EB split was not disclosed. [WDC-1; WDC-3]

Reported basis / period	Revenue	Gross profit	Operating expense	Operating profit
FY2026 Q3 adjusted; April 30 release	3,337	1,684	397	1,287
FY2026 Q4 GAAP; July 3 end	3,747	2,028	465	1,563

Reported basis / period	Revenue	Gross profit	Operating expense	Operating profit
FY2026 Q4 adjusted; July 3 end	3,747	2,037	382	1,655
Full FY2026 GAAP	12,919	6,311	1,858	4,453
Full FY2026 adjusted	12,919	6,349	1,532	4,817

The first forecast quarter is calendar Q3 2026, or fiscal Q1 2027. Management guidance is USD 4,000-4,200 million sales, 55-56% adjusted gross margin and USD 390-400 million adjusted operating expense. The model uses the midpoints: USD 4,100 million sales, approximately USD 2,276 million gross profit and USD 395 million expense, producing approximately USD 1,881 million adjusted operating profit. These remain estimates; the broker preview does not turn them into reported actuals. [WDC-1]

The guided sales allocation of USD 3,650 million cloud and USD 450 million client/consumer is analytical. Management guided consolidated sales rather than those exact product pools. An actual-quarter client/consumer margin proxy of 52% leaves a cloud margin near 55%; management described both categories as being in the 50s, without publishing that precise allocation. Exact product gross margins remain analytical assumptions.

Accounting basis, stock compensation and shares

The sanctioned earnings basis is adjusted operating profit. The actual-quarter GAAP-to-adjusted reconciliation adds USD 9 million to gross profit and removes USD 83 million from operating expense, increasing operating profit by USD 92 million. Gross-profit exclusions include USD 7 million stock compensation and USD 2 million other items. Expense exclusions include USD 38 million stock compensation, USD 42 million realignment and USD 3 million other items. Ordinary depreciation remains a production expense; operating profit is distinct from EBITDA. [WDC-1]

Total quarter stock compensation was USD 45 million. It is excluded under WDC's adjusted reporting convention, and remains a real equity/cash-replacement cost for investors. The model preserves that existing basis and does not treat it as free cash or claim a second earnings add-back. The next-quarter guidance excludes about USD 10 million from cost of sales and USD 60 million from operating expense. Future GAAP adjustments cannot be fully reconciled in advance. [WDC-1]

The quarter used 389 million GAAP diluted shares and 388 million adjusted diluted shares. The adjusted methodology considers capped calls against convertible dilution up to their limits. The next-quarter guide uses about 388 million shares and expects no material capped-call benefit. The model retains 388 million in every quarter, rather than substituting the lower dated basic share count. Future awards, repurchases and the scheduled convertible settlement must be reconciled to subsequent filings. [WDC-1; WDC-2; WDC-7]

The large gain on the retained Sandisk interest was non-operating. Its monetization and debt/equity transactions do not create recurring HDD sales or operating profit. The earlier retained Sandisk stake has been monetized; the forward perimeter contains no remaining flash earnings.

What the October pricing evidence supports

Morgan Stanley's October 20 preview attributes non-contracted nearline transactions to prices of USD 40-70 per TB, against approximately USD 15 blended. It also describes non-nearline sequential price increases of 15-30% or more, a broadening customer base and supply constraints in media and equipment. The stated industry shortfall of about 400 EB for 2027/28 was described as consistent with checks a month earlier. [HDD-P2, pages 2-3]

The report's 10-30% floating-sales example is hypothetical. The transaction sample, product/geographic mix, customer type, channel margin and net price reaching WDC remain undisclosed. Management's own calls describe base-price/base-volume contracts, separately priced upside volumes and staggered renewals. In August, management rejected an assumption of automatically accelerating contractual price escalators. Those mechanics constrain the amount and timing of producer capture. [WDC-3; WDC-4; WDC-5]

The selected case uses the low reported USD 40 endpoint. A 30% probability and 10% eligible shipped-EB share remain unchanged: this is a more recent observation from the same broker, without independent evidence of broad manufacturer realization. A higher conditional price is warranted within that bounded branch; a larger probability or exposed share would require a stronger transmission bridge. The source's upper endpoint is a diagnostic only.

are 10%+ above Street EPS forecasts in the December quarter (and think we could still be conservative in areas).

At the same time, we acknowledge the HDD catalyst path is cloudier today vs. a month ago – the market faces uncertainty on vendor rationality and sources of AI funding that won't be solved with earnings, and we have reduced our target multiples as a result. **But at ~8x P/E, current valuation implies HDD earnings peak in FY28 and that SSD pricing declines 80% over the next 2-3 years, severely limiting HDD pricing upside (more below).** Plausible? Sure. But (1) is that overly punitive given what we know about HDD, and broader AI, demand, and (2) does that create a positive asymmetrical setup where we can buy on oversold discounts? Yes, absolutely, in our view. As a result, we reiterate our Overweight rating on both STX ($1,200 PT; 82% upside) and WDC ($643 PT; 83% upside) and reinstate STX as our 'Top Pick' in IT Hardware.

First, let's get into the details of our recent checks – Toshiba, demand, pricing, etc. On Toshiba, our checks suggest there is little new HDD supply/demand risk – Toshiba remains gated by parts of the HDD supply chain (media, equipment), they are effectively already sold out of capacity for CY27, and recent expectations for a ~400EB HDD industry supply shortfall in CY27/CY28 (vs. 300EBs in CY26) are consistent with our checks a month ago. Also, the more work we do on Toshiba, the more it sounds like they are *more* aggressive on pricing than STX/WDC. Said differently, we remain comfortable with the Toshiba risk, despite the evolving headlines. On HDD demand and pricing – our checks make clear that EB demand has accelerated in recent months (recent cloud checks would support this), and that the customer base is broadening from core CSPs to new AI-centric customers (e.g. frontier labs) trying to buy HDD capacity in size (i.e. 100's of EBs). Supply shortages are limiting their ability to procure in size, but desperation for storage capacity is also pushing spot nearline HDD prices higher – we continue to hear about recent non-contracted deals getting closed at 4c/GB up to 7c/GB (vs. blended nearline pricing at ~1.5c/GB today). Finally, non-nearline HDD pricing remains extremely strong – and as NAND prices continue their march higher, the HDD industry is persistently raising non-nearline prices 15-30%+ Q/Q to capture better margins. It is our understanding all of these trends are consistent for both STX and WDC, although STX appears to be a bit more aggressive in capturing floating (e.g. spot) prices than WDC.

What does this mean for the STX/WDC earnings setup? A September quarter beat and a better-than expected pricing and profitability-driven raise in the December quarter. Somewhat controversially, we think this quarter's earnings setup is more attractive for WDC. Based on our investor conversations over the past month, we sense that investors are a bit less aggressive in modeling STX/WDC fundamentals, reflecting tempered enthusiasm following share underperformance over the past 90 days and recent Toshiba headlines (see here, here, and here). When juxtaposed versus the strength in our checks above, we believe the earnings setup for both WDC (Oct 22nd) and STX (Oct 27th) is attractive. And given the lower buyside bar at WDC, we think the F1Q earnings setup could be better for WDC over STX.

- **For STX:** For the September quarter, we forecast revenue of $4.16B (vs.

HDD-P2, original page 2. Morgan Stanley Research, October 20, 2026 at 03:44 UTC. Preserved private licensed exhibit. Transaction pricing, industry shortfall and broker opinions have different scopes; source analyst targets and probabilities are not model inputs.

Producer realization, non-nearline scope and preview timing

$4.00B-$4.20B guide) and a gross margin of 57.9%, leading to $7.50 of EPS vs. Street at $7.41. However, if recent history is a guide, results could come in closer to $4.26B of revenue and $8.13 of EPS. Looking to the December qtr, we model 6% Q/Q EB growth, MSD-HSD% Q/Q nearline pricing growth, higher-capacity mix and continued cost downs to drive $4.76B of revenue, a 62.1% gross margin, and EPS of $9.54, 6-12% above Consensus (Exhibit 1). However, these forecasts assume sequential pricing growth for nearline and non-nearline EBs decelerates from the September quarter, which would run counter to our checks. If we were to assume 10-30% of nearline bits are floating and subject to spot prices while 70-90% gets repriced at LSD-MSD% Q/Q rates in the Dec Q, that could deliver HSD%+ Q/Q nearline $/TB growth vs. our ~7% Q/Q forecast, with non-nearline pricing strength representing further upside to our estimates. Ultimately, this means our above-Street December quarter EPS forecast could still be 5%+ too low.

- **For WDC:** For September, we forecast revenue of $4.19B (vs. $4.00B-$4.20B guidance) and a gross margin of 56.0% (vs. 55-56% guidance), leading to $4.13 of EPS vs. Street at $4.09. The key swing factor in the Sept Q is nearline EB execution following last quarter's underperformance, particularly whether WDC delivers on the "pretty steep ramp" of 40TB ePMR expected in C2H26. Looking to December qtr, we model 8% Q/Q EB growth, MSD% Q/Q nearline pricing growth and improving higher-capacity mix to support revenue of $4.79B, a 59.5% gross margin, and EPS of $5.25, 5-10% above Consensus (Exhibit 2). Though using the same pricing analysis as we did for STX above, December qtr EPS could land 8% above our forecast, or closer to $5.67. For context, our investor conversations point to December quarter buyside expectations of $4.75B-$4.95B revenue, a 59.5%-61.0% gross margin, and $5.20-$5.60 of EPS. However, it is well-understood WDC guides relatively conservatively, and thus a guide at the low-to-middle-end of buyside estimates is expected.

Exhibit 1: STX: Our September quarter EPS estimate is at the high end of mgmt's guide and largely in line with Consensus, but our December quarter EPS estimate is 12% above Street, driven by stronger pricing and higher margin assumptions.

STX	Sep-26					Dec-26				
($M)	MSe New	MSe Old	Diff.	Cons	Diff.	MSe New	MSe Old	Diff.	Cons	Diff.
Revenue	4,163	4,168	0%	4,135	1%	4,755	4,740	0%	4,515	5%
Total HDD EB	*229*	*229*	*0%*	*237*	*-3%*	*243*	*249*	*-2%*	*249*	*-3%*
Nearline EB	*207*	*207*	*0%*	*213*	*-3%*	*219*	*225*	*-3%*	*222*	*-2%*
Total Price per EB	*$18.2*	*$18.2*	*0%*	*$17.5*	*4%*	*$19.6*	*$19.1*	*3%*	*$18.1*	*8%*
Total Cost per EB	*$7.7*	*$7.7*	*0%*	*$7.5*	*2%*	*$7.4*	*$7.3*	*1%*	*$7.5*	*0%*
Gross Profit	2,412	2,410	0%	2,360	2%	2,954	2,915	1%	2,666	11%
Operating Profits	2,108	2,108	0%	2,069	2%	2,650	2,612	1%	2,368	12%
Profit Before Tax	2,060	2,061	0%	2,013	2%	2,622	2,584	1%	2,320	13%
Net Income	1,731	1,731	0%	1,706	1%	2,202	2,171	1%	1,968	12%
EPS	**$7.50**	**$7.50**	**0%**	**$7.41**	**1%**	**$9.54**	**$9.39**	**2%**	**$8.54**	**12%**
Gross Margin	57.9%	57.8%	10bps	57.1%	90bps	62.1%	61.5%	60bps	58.8%	330bps
Operating Margin	50.6%	50.6%	10bps	50.0%	60bps	55.7%	55.1%	60bps	52.4%	330bps
Diluted Shares (M)	231	231	0%	230	0%	231	231	0%	230	0%

Source: Company data, FactSet, Morgan Stanley Research estimates

HDD-P2, original page 3. The 10-30% floating example is hypothetical. WDC September-quarter figures are broker forecasts ahead of October 22 results. Bank targets, earnings multiples and financial tables are not imported into this operating model. Private licensed source only.

The 400 EB shortfall does not create additional shippable WDC exabytes. Existing demand, manufacturing limits and qualification already represent the supply constraint. The source identifies no new SpaceX order or

accepted supplier allocation. Its bank targets, forward P/E choices and options-implied probabilities supply no additional valuation method or model weight.

Nearline probability, overlap and cost ledger

Ledger item	Selected treatment	Reason / counter
Signal and recognition	October 20 nearline noncontract price check; phased Q4 2026-Q3 2029	Same original broker/source group as earlier scarcity observation
Conditional endpoint	USD 40 per TB	Low end of reported USD 40-70 transactions; manufacturer realization unknown
Economic probability	30%	Protected contracts, staggered renewals, channel mix and producer pass-through remain uncertain
Eligible share	10% of qualified shipped EB	Analytical bound; source floating-share example is not an observed WDC mix
Reference already modeled	Remove the full reference price before weighting	The earlier USD 30 branch is replaced rather than stacked
Marginal cost	10% cost of sales plus 3% operating cost	87% marginal OP conversion; LOW cost proxy, not total product margin
Volume and qualification	Existing schedules retained	Shortage and broad AI demand add no separate shipments
Update/counter	Net producer price, contracts, accepted EB and component costs	Reduce transmission if spot quotes do not reach realized manufacturer sales

For each quarter, weighted price premium equals 30% x 10% x phase x the positive difference between USD 40 and the reference price. Qualified shipped EB times that premium gives the total selected scarcity revenue. Ordinary manufacturing cost, R&D and SG&A stay paid. Additional fulfillment/component/support cost consumes 10% of the rent and operating support another 3%.

The following table shows the TOTAL selected scarcity branch above reference pricing. It contains the contribution already present in the earlier branch. The separate revision table later isolates only the new change against the saved forecast.

Quarter	Phase	Shipped EB	Premium US cents/TB	Total branch R	COGS	Opex	Total branch OP
Q3 2026	0%	223	0	0	0	0	0
Q4 2026	25%	241	17	42	4	1	37
Q1 2027	50%	260	34	89	9	3	77
Q2 2027	75%	280	51	141	14	4	123
Q3 2027	100%	294	66	195	20	6	170

Quarter	Phase	Shipped EB	Premium US cents/TB	Total branch R	COGS	Opex	Total branch OP
Q4 2027	100%	312	66	205	20	6	178
Q1 2028	100%	327	66	214	21	6	186
Q2 2028	100%	342	66	224	22	7	195
Q3 2028	100%	364	66	238	24	7	207
Q4 2028	100%	373	66	245	24	7	213
Q1 2029	75%	405	49	199	20	6	173
Q2 2029	50%	425	33	140	14	4	122
Q3 2029	25%	440	16	73	7	2	63
Q4 2029	0%	440	0	0	0	0	0
Q1 2030	0%	488	0	0	0	0	0
Q2 2030	0%	510	0	0	0	0	0
Q3 2030	0%	533	0	0	0	0	0
Q4 2030	0%	540	0	0	0	0	0

The weighted premium is shown in whole US cents per TB; 100 cents equals USD 1. The exact premium and all pricing calculations are retained in the technical supplement. The USD 40 conditional endpoint is a transaction scenario. It does not become the company blended selling price.

WDC / OCTOBER 21, 2026 / FRAMEWORK 8.0

A separate, finite client and consumer price episode

Nearline USD per TB quotes do not transfer directly to finished client and consumer drives. Their units, retail/OEM split and net price exposure are unavailable. The selected LOW-confidence proxy starts from the existing USD 450 million Q3 2026 analytical pool at a fixed effective quantity. One 15% price step implies about USD 518 million conditional sales. The model subtracts the ENTIRE already-modeled client/consumer revenue in each eligible quarter before allowing a positive residual.

Only 25% of that residual is eligible, with 30% probability. The one-step observation applies from Q4 2026 through Q2 2027 and then expires. It is not compounded each quarter. Marginal component/manufacturing/support costs take 15% and operating costs 3%, leaving 82% of the bounded increment in adjusted operating profit. The larger cost reserve recognizes greater finished-product and channel scope. All of these eligibility and cost inputs are analytical assumptions.

Quarter	Conditional R	Existing R removed	Positive residual	Weighted new R	New COGS	New opex	New OP
Q4 2026	518	480	38	3	<1	<1	2

Quarter	Conditional R	Existing R removed	Positive residual	Weighted new R	New COGS	New opex	New OP
Q1 2027	518	475	43	3	<1	<1	3
Q2 2027	518	490	28	2	<1	<1	2

The three-quarter episode adds about USD 8 million sales and USD 7 million adjusted operating profit after costs. A 10% lower effective-quantity proxy leaves no positive residual after existing growth; a 10% higher proxy leaves a larger residual. Those bounds measure unknown mix/quantity overlap, without adding physical shipments. The episode ends before the active Q4 2027 earnings window, so it contributes zero to that window for a timing reason.

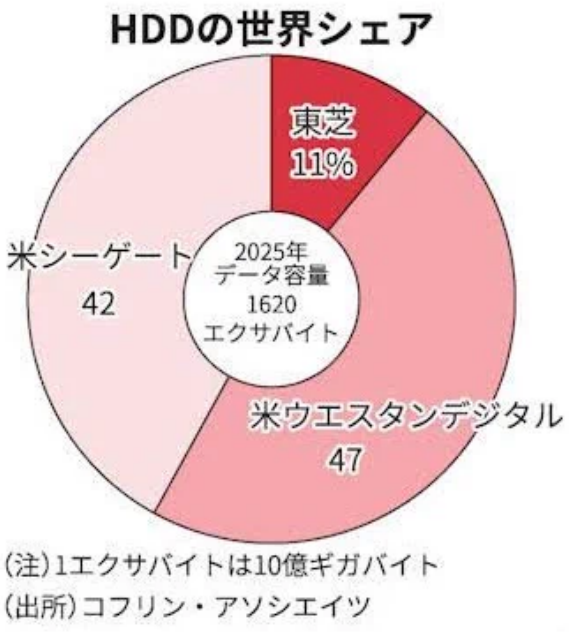
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Toshiba competition remains a paid downside

Toshiba's October 2 release confirms first shipments from the expanded Laguna Technopark nearline line. Its target is nearly twice FY2025 annual production on a storage-capacity basis by FY2027, which ends March 2028. Bytes, finished-drive units and revenue are different denominators. Exact accepted EB and utilization are undisclosed. [TOS-1]

The model retains a 50% probability of a 4% price concession on 50% of repriceable cloud sales. Its ramp starts at 25% in Q4 2027, then 50%, 75% and full from Q3 2028. From 2029, 50% overlap is removed against normalization already in reference pricing. This is an absolute dollar/price loss on the reference pool; it is not enlarged by the new scarcity premium. No production-cost saving is assumed from a price concession.

The broker's sold-out, constrained-media and pricing-rationality checks are useful counter-evidence. They do not erase the issuer's first-shipment milestone or establish that later qualified supply cannot affect renewals. The downside and scarcity branches remain separately scoped and enter earnings once.



Inherited Nikkei / Techno Systems Research exhibit: 2025 HDD share by storage capacity, not units or revenue. Toshiba 11%, WDC 47%, Seagate 42%; total 1,620 EB. This historical mix does not quantify new factory output.

Quarter	Reference USD/TB	Toshiba price loss US cents/TB	Toshiba OP loss	Active USD/TB
Q3 2026	16	0	0	16
Q4 2026	17	0	0	17
Q1 2027	17	0	0	18
Q2 2027	18	0	0	18
Q3 2027	18	0	0	19
Q4 2027	18	5	-14	19
Q1 2028	18	9	-30	19
Q2 2028	18	14	-47	19
Q3 2028	18	18	-66	19
Q4 2028	18	18	-68	19
Q1 2029	18	9	-37	19
Q2 2029	18	9	-38	18
Q3 2029	18	9	-40	18
Q4 2029	18	9	-40	18
Q1 2030	18	9	-44	18
Q2 2030	18	9	-45	18

Quarter	Reference USD/TB	Toshiba price loss US cents/TB	Toshiba OP loss	Active USD/TB
Q3 2030	18	9	-47	18
Q4 2030	18	9	-48	18

The small Toshiba price concession is shown in whole US cents per TB, while reference and active prices remain USD per TB. Its exact effect remains present in the model, signed OP loss and technical supplement.

A calibrated storage-demand bridge

The retained customer-spending basket is a demand index: Amazon, Microsoft, Alphabet and Meta, with 2026 reference spending of USD 220,000 million, USD 175,000 million, USD 200,000 million and USD 137,500 million. Definitions differ across cash capital expenditure, lease classification and finance-lease principal. These are neither HDD supplier revenue nor disclosed WDC customer allocations. [D1-D4]

The reference basket totals USD 732,500 million. Calibrating USD 3,335 million reported cloud sales against one quarter of that basket gives about a 2% effective supplier coefficient. That single-point coefficient also embeds other cloud/enterprise customers; it does not prove individual customer shares. The physical bridge uses 90% of reported 231 total EB, or about 208 cloud EB, as a LOW-confidence calibration proxy. Exact coefficient and EB are retained in the supplement.

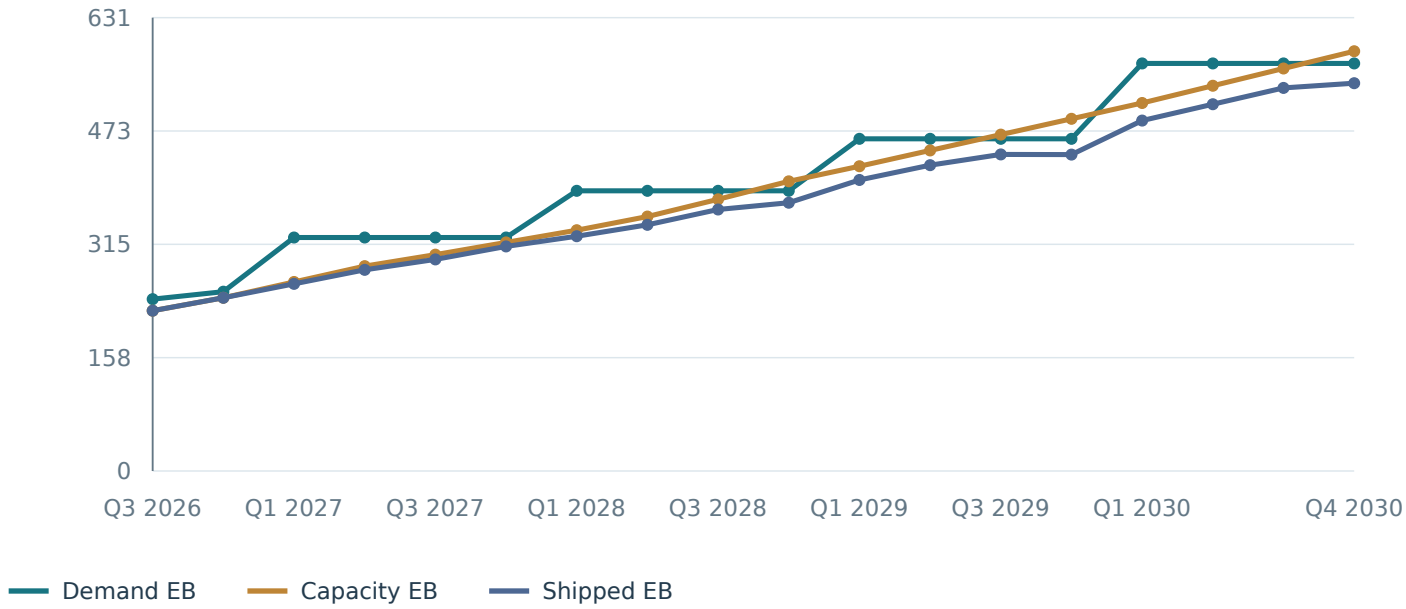
The existing spending index grows 25% in 2027, 20% in 2028, 14% in 2029 and 10% in 2030. Retained-data intensity is separately assumed, reaching a 45% uplift to the anchor by 2030. Demand EB is capped by physical capacity before applying the HAMR qualification reserve. Storage sales then depend on shipped EB and price. The same AI spending and persistent-data story is not added again as another revenue pool.

Funding, power availability and data-center construction can delay the spending-to-deployment connection. The October report offers no new WDC order, wallet share or accepted customer volume from SpaceX. Existing capex-driven demand therefore receives no second macro uplift. NAND price strength can aid HDD value in some applications while increasing DRAM/NAND input expense inside the drives.

Demand, manufacturing capacity and accepted shipments

Physical cloud storage: demand, capacity and qualified shipments

Exabytes; all observations are analytical forecast quantities



Potential EB is the lesser of demand and capacity. Planned HAMR receives its qualification/delivery factor once; missed HAMR EB are not automatically replaced. Capacity and intensity are LOW proxies beyond the near-term guidance anchor.

Quarter	Demand EB	Capacity EB	Shipped EB
Q3 2026	239	223	223
Q4 2026	249	241	241
Q1 2027	325	263	260
Q2 2027	325	285	280
Q3 2027	325	301	294
Q4 2027	325	318	312
Q1 2028	390	335	327
Q2 2028	390	354	342
Q3 2028	390	378	364
Q4 2028	390	403	373
Q1 2029	462	424	405

Quarter	Demand EB	Capacity EB	Shipped EB
Q2 2029	462	446	425
Q3 2029	462	468	440
Q4 2029	462	490	440
Q1 2030	567	512	488
Q2 2030	567	536	510
Q3 2030	567	560	533
Q4 2030	567	584	540

The capacity path rises from 223 EB in Q3 2026 to 318 EB in Q4 2027 and 584 EB in Q4 2030. Density and automation support that analytical path; it is not disclosed factory-output guidance. It carries no separate second benefit from Toshiba's or the industry's stated exabyte capacity.

Qualification, acceptance and product substitution

40TB ePMR was in volume ramp with two customers at the August call. Management targeted at least half of nearline exabytes on that platform by fiscal Q3 2027. UltraSMR was expected to approach 60% by fiscal year-end, and may overlap ePMR or HAMR. These labels do not define additive revenue pools. [WDC-3]

September management described initial small HAMR shipments in Q1 2027, more in Q2 and a broader second-half ramp. The model applies 80%, 85%, 90% and then 95% conversion to planned HAMR EB, and zero HAMR revenue in 2026. Those numerical weights are analytical. No company guidance for year-end HAMR share was given. High-bandwidth samples at five customers and a roadmap alongside 50TB-plus drives have no separate revenue add. [WDC-3; WDC-5]

Quarter	Planned HAMR mix	HAMR conversion	ePMR mix	HAMR EB	ePMR EB	Other EB
Q3 2026	0%	100%	25%	0	56	167
Q4 2026	0%	100%	40%	0	96	145
Q1 2027	5%	80%	50%	11	132	118
Q2 2027	12%	85%	60%	29	171	80
Q3 2027	22%	90%	55%	60	166	69
Q4 2027	35%	95%	45%	106	143	64
Q1 2028	50%	95%	35%	159	117	50
Q2 2028	65%	95%	25%	219	89	35

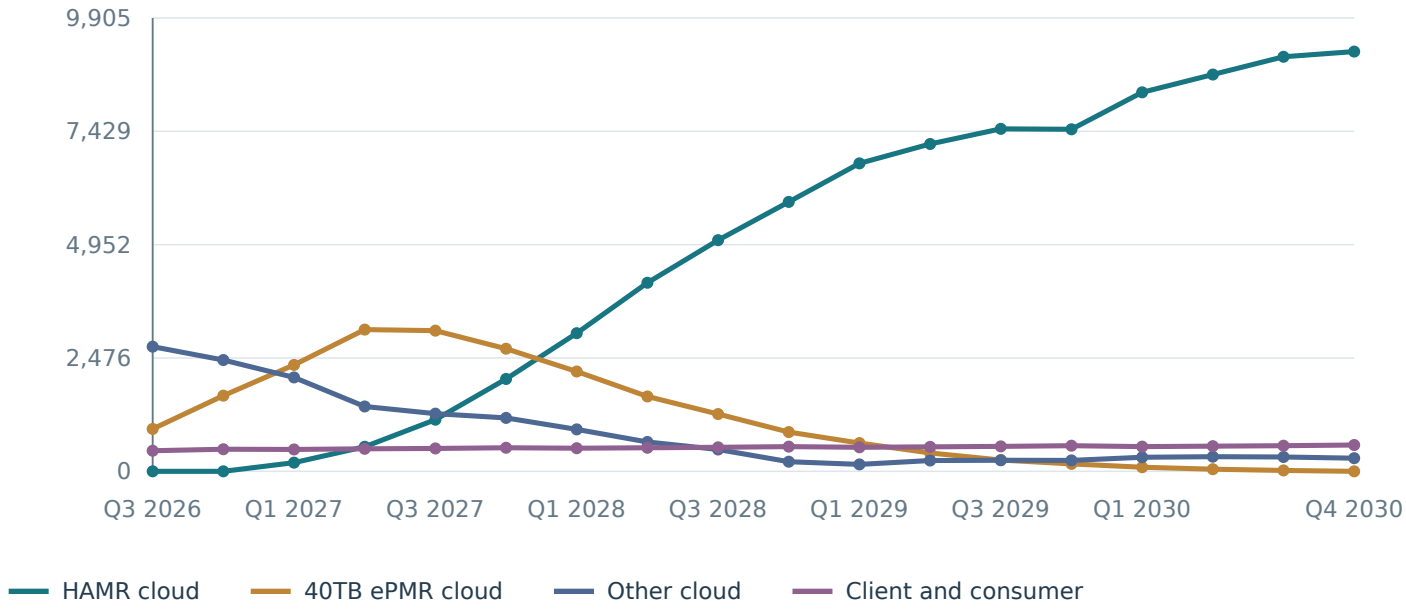
Quarter	Planned HAMR mix	HAMR conversion	ePMR mix	HAMR EB	ePMR EB	Other EB
Q3 2028	75%	95%	18%	269	68	26
Q4 2028	85%	95%	12%	315	47	12
Q1 2029	90%	95%	8%	363	34	8
Q2 2029	92%	95%	5%	390	22	13
Q3 2029	94%	95%	3%	413	14	14
Q4 2029	95%	95%	2%	417	9	14
Q1 2030	96%	95%	1%	465	5	18
Q2 2030	96%	95%	<1%	489	3	19
Q3 2030	97%	95%	<1%	513	1	18
Q4 2030	97%	95%	0%	523	0	17

The later mix is LOW confidence. A qualification miss reduces the affected HAMR revenue and associated units once; it does not justify multiplying the entire company valuation by the same probability again. Other production remains on the retained plan.

Product sales and gross-profit architecture

Model revenue by product pool

USD millions; forecast product allocations, not reported segment guidance



HAMR replaces part of ePMR and older cloud drives. Relative price premiums of 4% for HAMR and 2% for advanced ePMR are normalized to the selected blended price; they are analytical assumptions. Client/consumer is a separate combined pool.

Quarter	HAMR cloud	40TB ePMR cloud	Other cloud	Client and consumer
Q3 2026	0	926	2,724	450
Q4 2026	0	1,654	2,432	483
Q1 2027	190	2,325	2,052	478
Q2 2027	537	3,097	1,417	492
Q3 2027	1,128	3,074	1,260	500
Q4 2027	2,018	2,679	1,167	515
Q1 2028	3,018	2,181	916	505
Q2 2028	4,120	1,636	642	515
Q3 2028	5,051	1,251	477	525
Q4 2028	5,886	858	210	540
Q1 2029	6,729	617	151	525
Q2 2029	7,152	401	236	535

Quarter	HAMR cloud	40TB ePMR cloud	Other cloud	Client and consumer
Q3 2029	7,483	247	242	545
Q4 2029	7,473	162	239	560
Q1 2030	8,281	90	307	540
Q2 2030	8,669	47	320	550
Q3 2030	9,058	19	314	560
Q4 2030	9,171	0	287	575

Each product table separates retained manufacturing cost from the marginal cost of pricing. The retained cost includes ordinary production depreciation through the original margin schedules. The inherited model has no independently observed fixed/variable manufacturing split or asset-activation cohorts. A more detailed cash/D&A engine cannot be inferred from these product margins.

Reference HAMR gross margin reaches 70%; advanced ePMR peaks around 65% and other cloud settles around 53%. These are long-dated LOW-confidence cost and mix assumptions. Realized margins depend on yield, memory inputs, reliability, production expense and customer value-sharing. The new price check does not prove those future costs or a permanent monopoly.

HAMR cloud drives - all operating quarters

Quarter	Revenue	Retained factory COGS	Pricing COGS	Total COGS	Gross profit	Gross margin
Q3 2026	0	0	0	0	0	-
Q4 2026	0	0	0	0	0	-
Q1 2027	190	76	<1	77	113	60%
Q2 2027	537	203	2	205	332	62%
Q3 2027	1,128	397	4	401	727	64%
Q4 2027	2,018	674	7	681	1,337	66%
Q1 2028	3,018	981	11	991	2,027	67%
Q2 2028	4,120	1,302	14	1,316	2,804	68%
Q3 2028	5,051	1,551	18	1,568	3,482	69%
Q4 2028	5,886	1,750	21	1,770	4,116	70%
Q1 2029	6,729	1,975	18	1,993	4,736	70%
Q2 2029	7,152	2,118	13	2,130	5,022	70%
Q3 2029	7,483	2,236	7	2,242	5,240	70%
Q4 2029	7,473	2,253	0	2,253	5,220	70%
Q1 2030	8,281	2,497	0	2,497	5,784	70%
Q2 2030	8,669	2,614	0	2,614	6,056	70%
Q3 2030	9,058	2,731	0	2,731	6,327	70%
Q4 2030	9,171	2,765	0	2,765	6,406	70%

USD millions; analytical product allocation. Pricing COGS represents the marginal manufacturing/component/fulfillment reserve, in addition to retained factory cost. A zero-revenue pool has no meaningful calculated gross margin. This is distinct from company-reported segment profitability.

40TB-class ePMR cloud drives - all operating quarters

Quarter	Revenue	Retained factory COGS	Pricing COGS	Total COGS	Gross profit	Gross margin
Q3 2026	926	389	0	389	537	58%
Q4 2026	1,654	671	2	673	981	59%
Q1 2027	2,325	901	5	905	1,420	61%
Q2 2027	3,097	1,144	9	1,153	1,944	63%
Q3 2027	3,074	1,097	11	1,108	1,966	64%
Q4 2027	2,679	933	9	942	1,736	65%
Q1 2028	2,181	751	8	759	1,423	65%
Q2 2028	1,636	565	6	570	1,066	65%
Q3 2028	1,251	439	4	443	808	65%
Q4 2028	858	301	3	304	554	65%
Q1 2029	617	220	2	222	395	64%
Q2 2029	401	145	<1	145	256	64%
Q3 2029	247	91	<1	91	155	63%
Q4 2029	162	60	0	60	102	63%
Q1 2030	90	33	0	33	56	63%
Q2 2030	47	17	0	17	29	63%
Q3 2030	19	7	0	7	12	63%
Q4 2030	0	0	0	0	0	-

USD millions; analytical product allocation. Pricing COGS represents the marginal manufacturing/component/fulfillment reserve, in addition to retained factory cost. A zero-revenue pool has no meaningful calculated gross margin. This is distinct from company-reported segment profitability.

Other cloud drives - all operating quarters

Quarter	Revenue	Retained factory COGS	Pricing COGS	Total COGS	Gross profit	Gross margin
Q3 2026	2,724	1,229	0	1,229	1,495	55%
Q4 2026	2,432	1,071	2	1,073	1,358	56%
Q1 2027	2,052	885	4	889	1,163	57%
Q2 2027	1,417	606	4	610	807	57%
Q3 2027	1,260	541	5	545	715	57%
Q4 2027	1,167	508	4	512	655	56%
Q1 2028	916	404	3	408	509	56%
Q2 2028	642	287	2	289	352	55%
Q3 2028	477	216	2	218	259	54%
Q4 2028	210	96	<1	97	113	54%
Q1 2029	151	70	<1	71	81	53%
Q2 2029	236	111	<1	111	125	53%
Q3 2029	242	114	<1	115	127	53%
Q4 2029	239	114	0	114	125	52%
Q1 2030	307	147	0	147	161	52%
Q2 2030	320	153	0	153	167	52%
Q3 2030	314	150	0	150	164	52%
Q4 2030	287	137	0	137	150	52%

USD millions; analytical product allocation. Pricing COGS represents the marginal manufacturing/component/fulfillment reserve, in addition to retained factory cost. A zero-revenue pool has no meaningful calculated gross margin. This is distinct from company-reported segment profitability.

Client and consumer HDDs - all operating quarters

Quarter	Revenue	Retained factory COGS	Pricing COGS	Total COGS	Gross profit	Gross margin
Q3 2026	450	207	0	207	243	54%
Q4 2026	483	216	<1	216	266	55%
Q1 2027	478	211	<1	212	266	56%
Q2 2027	492	216	<1	216	276	56%
Q3 2027	500	218	0	218	283	57%
Q4 2027	515	221	0	221	294	57%
Q1 2028	505	217	0	217	288	57%
Q2 2028	515	219	0	219	296	58%
Q3 2028	525	223	0	223	302	58%
Q4 2028	540	227	0	227	313	58%
Q1 2029	525	223	0	223	302	58%
Q2 2029	535	227	0	227	308	58%
Q3 2029	545	234	0	234	311	57%
Q4 2029	560	241	0	241	319	57%
Q1 2030	540	232	0	232	308	57%
Q2 2030	550	237	0	237	314	57%
Q3 2030	560	241	0	241	319	57%
Q4 2030	575	247	0	247	328	57%

USD millions; analytical product allocation. Pricing COGS represents the marginal manufacturing/component/fulfillment reserve, in addition to retained factory cost. A zero-revenue pool has no meaningful calculated gross margin. This is distinct from company-reported segment profitability.

Consolidated operating bridge

Reference R&D and SG&A remain paid in dollars, using the inherited 67% / 33% allocation of central expense. Reference operating expense rises from USD 395 million to USD 516 million across the forecast. The pricing branches separately pay their 3% operating support charge. Neither a price increase nor a demand headline automatically reduces staffing, engineering or ordinary manufacturing expense.

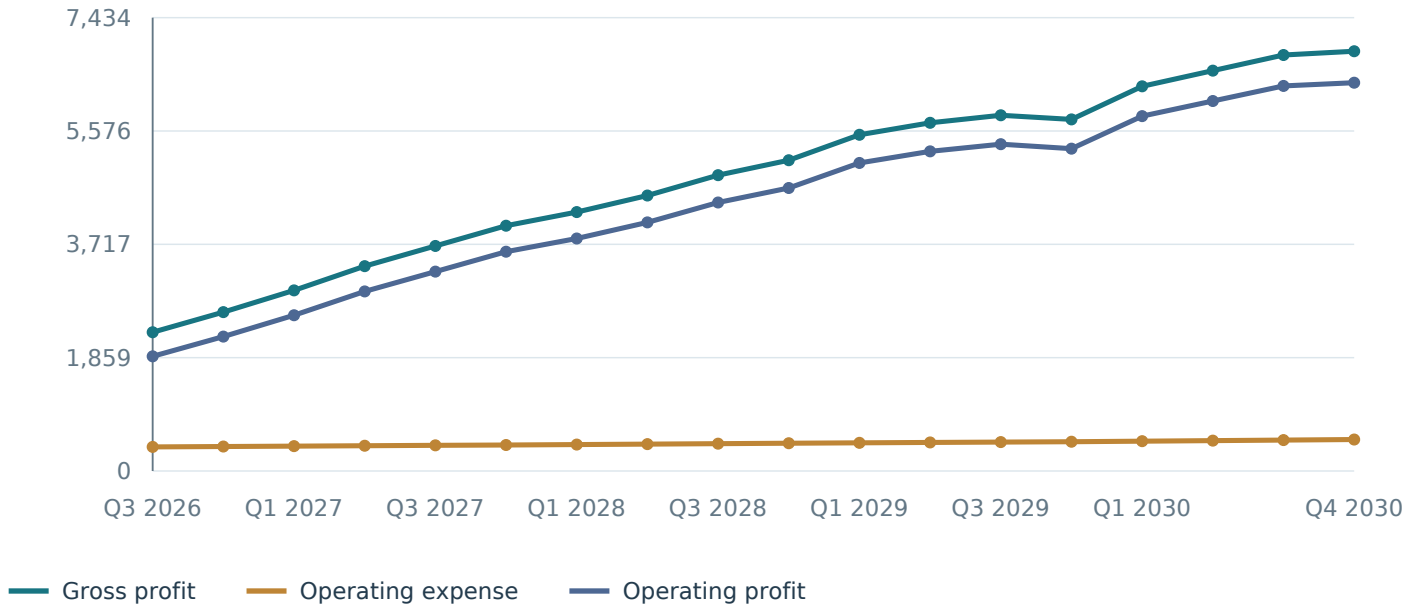
Quarter	Revenue	COGS	Gross profit	R&D	SG&A	Other opex	Operating profit	OP margin
Q3 2026	4,100	1,825	2,276	265	130	0	1,881	46%
Q4 2026	4,568	1,963	2,605	268	132	1	2,204	48%
Q1 2027	5,045	2,083	2,962	271	134	3	2,554	51%
Q2 2027	5,542	2,183	3,359	275	135	4	2,945	53%
Q3 2027	5,962	2,271	3,691	278	137	6	3,270	55%
Q4 2027	6,379	2,357	4,022	281	139	6	3,596	56%
Q1 2028	6,621	2,375	4,246	286	141	6	3,813	58%
Q2 2028	6,912	2,395	4,518	291	143	7	4,077	59%
Q3 2028	7,304	2,453	4,851	295	146	7	4,403	60%
Q4 2028	7,495	2,398	5,096	300	148	7	4,641	62%
Q1 2029	8,022	2,509	5,514	306	150	6	5,052	63%
Q2 2029	8,324	2,614	5,710	311	153	4	5,242	63%
Q3 2029	8,516	2,682	5,834	316	156	2	5,359	63%
Q4 2029	8,434	2,668	5,766	322	158	0	5,286	63%
Q1 2030	9,217	2,909	6,309	328	161	0	5,820	63%
Q2 2030	9,586	3,020	6,566	334	164	0	6,068	63%
Q3 2030	9,951	3,129	6,822	340	167	0	6,315	63%
Q4 2030	10,033	3,149	6,884	346	170	0	6,368	63%

All rows are forecasts on the existing adjusted operating basis. The gross-profit adjustment is zero throughout. Stock-compensation exclusions follow the sanctioned adjusted method; ordinary depreciation remains in cost. Later operating margins are particularly sensitive to the assumed HAMR mix, yields and paid cost path.

Profit after ordinary manufacturing and operating costs

Gross profit, operating expense and operating profit

USD millions; adjusted-basis model forecasts



Operating profit follows gross profit less all modeled operating expenses. High later margins depend on LOW-confidence product/cost and qualification assumptions; no GAAP or free-cash-flow equivalence is implied.

Quarter	Gross profit	Operating expense	Operating profit
Q3 2026	2,276	395	1,881
Q4 2026	2,605	401	2,204
Q1 2027	2,962	408	2,554
Q2 2027	3,359	414	2,945
Q3 2027	3,691	421	3,270
Q4 2027	4,022	426	3,596
Q1 2028	4,246	433	3,813
Q2 2028	4,518	441	4,077
Q3 2028	4,851	448	4,403
Q4 2028	5,096	455	4,641
Q1 2029	5,514	462	5,052

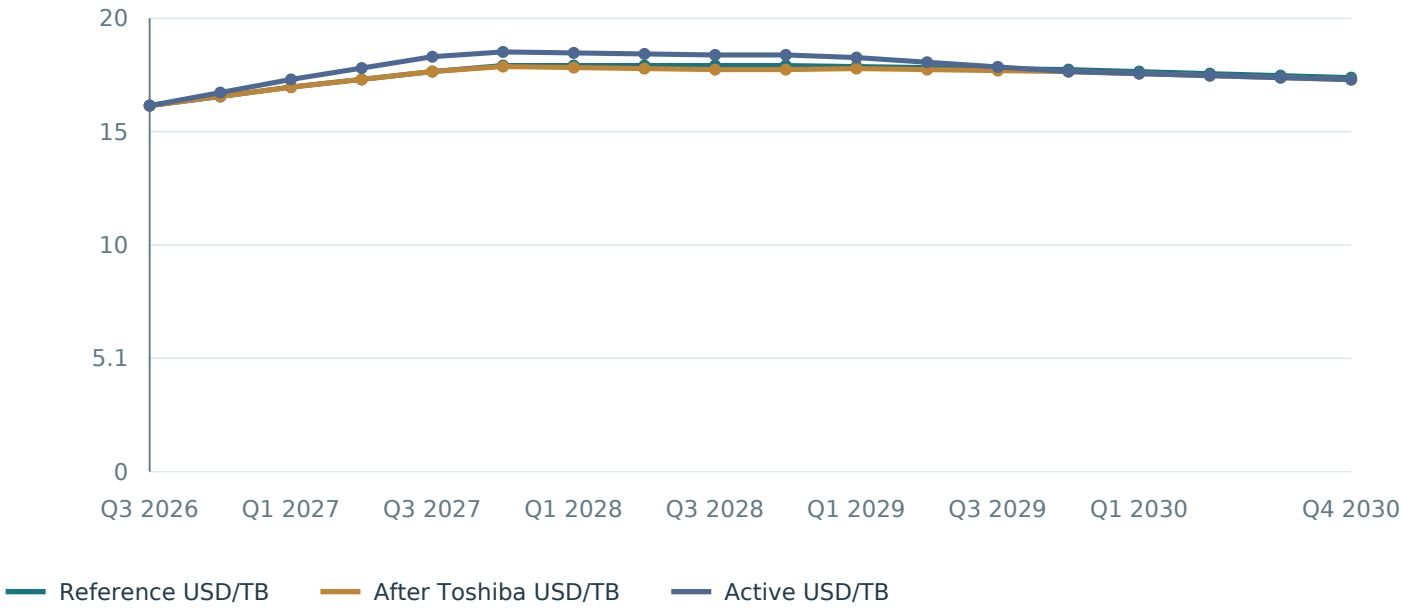
Quarter	Gross profit	Operating expense	Operating profit
Q2 2029	5,710	468	5,242
Q3 2029	5,834	474	5,359
Q4 2029	5,766	480	5,286
Q1 2030	6,309	489	5,820
Q2 2030	6,566	498	6,068
Q3 2030	6,822	507	6,315
Q4 2030	6,884	516	6,368

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Pricing path and finite fade

Qualified cloud pricing after scarcity and competitor effects

USD per TB; analytical blended prices, whole-number reader table



The conditional USD 40 transaction endpoint is applied only through probability, exposure and phase. It is not the active blended price. The scarcity branch fades to zero by Q4 2029, while the retained Toshiba concession continues.

Quarter	Reference USD/TB	After Toshiba USD/TB	Active USD/TB
Q3 2026	16	16	16
Q4 2026	17	17	17
Q1 2027	17	17	18

Quarter	Reference USD/TB	After Toshiba USD/TB	Active USD/TB
Q2 2027	18	18	18
Q3 2027	18	18	19
Q4 2027	18	18	19
Q1 2028	18	18	19
Q2 2028	18	18	19
Q3 2028	18	18	19
Q4 2028	18	18	19
Q1 2029	18	18	19
Q2 2029	18	18	18
Q3 2029	18	18	18
Q4 2029	18	18	18
Q1 2030	18	18	18
Q2 2030	18	18	18
Q3 2030	18	18	18
Q4 2030	18	18	18

Reference pricing already rises through 2027, holds in 2028 and eases in 2029-30. The scarcity phase is 25%, 50%, 75% and then full from Q4 2026 through Q4 2028; it falls to 75%, 50%, 25% and zero during 2029. The finite branch avoids capitalizing one urgent spot-market episode forever.

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Incremental revision versus the saved forecast

The table below isolates the new revision against the saved forecast at identical quarters, currency, multiple and share basis. This is different from the total scarcity branch above. Existing pricing upside and all previously paid costs remain in the reference model. No horizon roll contributes to the revision.

Quarter	New sales vs before	New COGS vs before	New opex vs before	New OP vs before	Client part of new OP
Q3 2026	0	0	0	0	0
Q4 2026	21	2	<1	18	2
Q1 2027	42	4	1	37	3
Q2 2027	65	7	2	56	2

Quarter	New sales vs before	New COGS vs before	New opex vs before	New OP vs before	Client part of new OP
Q3 2027	88	9	3	77	0
Q4 2027	94	9	3	82	0
Q1 2028	98	10	3	85	0
Q2 2028	103	10	3	89	0
Q3 2028	109	11	3	95	0
Q4 2028	112	11	3	97	0
Q1 2029	91	9	3	79	0
Q2 2029	64	6	2	56	0
Q3 2029	33	3	<1	29	0
Q4 2029	0	0	0	0	0
Q1 2030	0	0	0	0	0
Q2 2030	0	0	0	0	0
Q3 2030	0	0	0	0	0
Q4 2030	0	0	0	0	0

Within the active Q4 2027-through-Q3 2028 earnings window, the replacement price endpoint adds approximately USD 404 million sales and USD 351 million adjusted operating profit after marginal cost. The client episode has already expired. The same-window model ceiling rises from about USD 801 to USD 819, or about 2%. Exact comparisons remain in the precision supplement.

Calendar and fiscal windows

Calendar buckets map to the closest fiscal quarter: Q3 2026 is fiscal Q1 2027; Q4 is fiscal Q2; Q1 2027 is fiscal Q3; Q2 is fiscal Q4. Fiscal years end around June. The reported July 3 actual belongs to the preceding June/calendar Q2 bucket. Forecast Q3 2026 must not be presented as an already reported quarter.

Calendar year	Forecast coverage	Revenue	Gross profit	Operating profit
2026	Two quarters	8,668	4,881	4,085
2027	Four quarters	22,928	14,034	12,365
2028	Four quarters	28,332	18,712	16,934
2029	Four quarters	33,297	22,824	20,939
2030	Four quarters	38,787	26,580	24,570

Fiscal year ending around June	Operating quarters	Revenue	Operating profit
2027	Q3 2026 through Q2 2027	19,255	9,584
2028	Q3 2027 through Q2 2028	25,874	14,756
2029	Q3 2028 through Q2 2029	31,146	19,338
2030	Q3 2029 through Q2 2030	35,754	22,533

FY2027 is the first four modeled quarters. WDC reported USD 12,919 million FY2026 revenue but did not provide a complete full-year dollar forecast in the latest call. Annual aggregates above are the model, constrained by its supply, qualification and pricing assumptions.

Cash, capital expenditure and financing limits

At July 3, cash was USD 1,579 million and current debt USD 1,052 million. Those dated balances are not a current cash snapshot. Quarter operating cash flow was USD 1,389 million, purchases of property and equipment net were USD 108 million, and free cash flow USD 1,281 million under WDC's definition. Quarter depreciation and amortization was USD 99 million. [WDC-1; WDC-2]

On September 14, WDC called about USD 110 million remaining principal of its 3% convertible notes for November 16 redemption. The settlement was still future at this review date. No new retirement, cash distribution, lower diluted denominator or debt-funded growth benefit is assumed. Exact principal is retained in the supplement. The reported quarter included convertible settlement and repurchase transactions; those cash uses are not recurring operating-profit add-backs. [WDC-7]

September management said head/media and automation spending could run slightly above the long-term 4-6% of revenue range for the next couple of years. The existing cash cross-check uses 7% of revenue capex in 2027, approximately 7% in 2028, 6% in 2029 and approximately 6% in 2030 when rounded; exact assumptions are in the supplement. These are analytical allowances, not company capital guidance. [WDC-5]

No complete quarterly forward cash, debt maturity, lease principal, working-capital, award-settlement or commissioned-asset schedule is available in this inherited model. Its adjusted operating-profit ceiling therefore does not demonstrate that all future capital needs are funded or that profit converts one-for-one into distributable cash. Higher memory input expense, yields, equipment timing or refinancing costs could reduce those economics. No new dilution or financing is invented from the pricing report.

Seven-category analytical reassessment

Category	Points	Current reassessment
Core Thesis	17	Scaled, qualified head/media and reliability platform with few alternatives. Higher prices improve rent capture; Seagate and Toshiba prevent monopoly credit.
Cycle Stage	9	40TB ramp and scarcity remain early-to-mid commercialization evidence. Upcoming earnings and HAMR customer acceptance remain unresolved.
TAM Expansion	8	Persistent cloud/AI data supports the existing storage wallet. No new supplier-specific award or accepted exabytes is demonstrated.
Catalysts	5	October 22 results, 2027 renewals, 40TB adoption and HAMR acceptance remain measurable triggers. The bank preview completes none of them.
Macro	9	Data growth and cloud investment remain strong. Funding, power/construction and input-memory costs can affect timing and returns.
Alpha	2	Widely circulated bank checks improve calibration. The edge requires manufacturer realization, qualification and cost analysis rather than exclusive source access.
Risks	-4	Four distinct classes: technology/yield; demand and customer concentration; competition/substitution; financial/input-cost/policy. Each deducted once.
Total	46 / 60 = 77 / 100	High Conviction under the existing scale; independent of current price and valuation multiple

All seven categories were reassessed. The pricing revision strengthens the modeled earnings input without proving a new moat, exclusive alpha, completed catalyst or funded expansion. Analytical Base remains 77, and the protected portfolio Excel Base separately remains 77. Neither score is an additional probability haircut on operating profit or the ceiling.

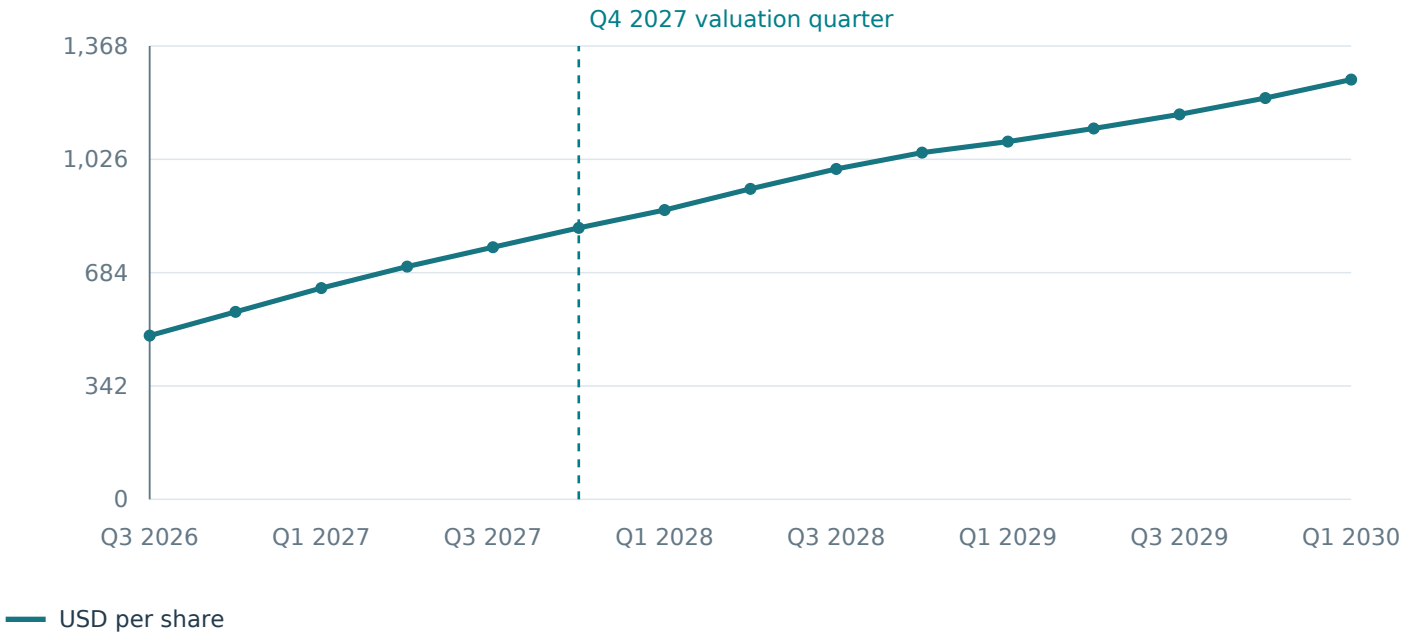
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Fifteen complete operating-profit valuations

The owner-selected method is 20x adjusted operating profit for the current valuation quarter plus the following three quarters, divided by that quarter's diluted shares. The denominator is 388 million throughout. This is the existing operating-profit valuation convention; no bank P/E, net-income conversion, enterprise-value adjustment or second probability is substituted. Longer-dated ceilings are conditional and carry more uncertainty.

All fifteen valuation-quarter ceilings

USD per share; 20x forward four-quarter adjusted operating profit / 388 million diluted shares



The highlighted Q4 2027 valuation uses Q4 2027 through Q3 2028 operating profit. Every observation is derived from the four explicit terms below; no future-price assumption changes shares.

Quarter	USD per share
Q3 2026	494
Q4 2026	566
Q1 2027	637
Q2 2027	702
Q3 2027	761
Q4 2027	819
Q1 2028	873
Q2 2028	937
Q3 2028	997
Q4 2028	1,046
Q1 2029	1,079
Q2 2029	1,119
Q3 2029	1,161
Q4 2029	1,211
Q1 2030	1,266

Ceiling quarter	Operating earnings window	Four-quarter OP	Multiple	Shares M	Ceiling USD
Q3 2026	Q3 2026 - Q2 2027	9,584	20x	388	494
Q4 2026	Q4 2026 - Q3 2027	10,973	20x	388	566
Q1 2027	Q1 2027 - Q4 2027	12,365	20x	388	637
Q2 2027	Q2 2027 - Q1 2028	13,624	20x	388	702
Q3 2027	Q3 2027 - Q2 2028	14,756	20x	388	761
Q4 2027	Q4 2027 - Q3 2028	15,889	20x	388	819
Q1 2028	Q1 2028 - Q4 2028	16,934	20x	388	873
Q2 2028	Q2 2028 - Q1 2029	18,173	20x	388	937
Q3 2028	Q3 2028 - Q2 2029	19,338	20x	388	997
Q4 2028	Q4 2028 - Q3 2029	20,294	20x	388	1,046
Q1 2029	Q1 2029 - Q4 2029	20,939	20x	388	1,079
Q2 2029	Q2 2029 - Q1 2030	21,707	20x	388	1,119
Q3 2029	Q3 2029 - Q2 2030	22,533	20x	388	1,161
Q4 2029	Q4 2029 - Q3 2030	23,488	20x	388	1,211
Q1 2030	Q1 2030 - Q4 2030	24,570	20x	388	1,266

Explicit earnings chains 1-3

Q3 2026 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q3 2026	1,881
2	Q4 2026	2,204
3	Q1 2027	2,554
4	Q2 2027	2,945
Sum	Forward four quarters	9,584
Valuation	Sum x 20 / 388 million shares	USD 494 per share

Q4 2026 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q4 2026	2,204
2	Q1 2027	2,554
3	Q2 2027	2,945
4	Q3 2027	3,270
Sum	Forward four quarters	10,973
Valuation	Sum x 20 / 388 million shares	USD 566 per share

Q1 2027 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q1 2027	2,554
2	Q2 2027	2,945
3	Q3 2027	3,270
4	Q4 2027	3,596
Sum	Forward four quarters	12,365
Valuation	Sum x 20 / 388 million shares	USD 637 per share

Explicit earnings chains 4-6

Q2 2027 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q2 2027	2,945
2	Q3 2027	3,270
3	Q4 2027	3,596
4	Q1 2028	3,813
Sum	Forward four quarters	13,624
Valuation	Sum x 20 / 388 million shares	USD 702 per share

Q3 2027 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q3 2027	3,270
2	Q4 2027	3,596
3	Q1 2028	3,813
4	Q2 2028	4,077
Sum	Forward four quarters	14,756
Valuation	Sum x 20 / 388 million shares	USD 761 per share

Q4 2027 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q4 2027	3,596
2	Q1 2028	3,813
3	Q2 2028	4,077
4	Q3 2028	4,403
Sum	Forward four quarters	15,889
Valuation	Sum x 20 / 388 million shares	USD 819 per share

Explicit earnings chains 7-9

Q1 2028 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q1 2028	3,813
2	Q2 2028	4,077
3	Q3 2028	4,403
4	Q4 2028	4,641
Sum	Forward four quarters	16,934
Valuation	Sum x 20 / 388 million shares	USD 873 per share

Q2 2028 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q2 2028	4,077
2	Q3 2028	4,403
3	Q4 2028	4,641
4	Q1 2029	5,052
Sum	Forward four quarters	18,173
Valuation	Sum x 20 / 388 million shares	USD 937 per share

Q3 2028 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q3 2028	4,403
2	Q4 2028	4,641
3	Q1 2029	5,052
4	Q2 2029	5,242
Sum	Forward four quarters	19,338
Valuation	Sum x 20 / 388 million shares	USD 997 per share

Explicit earnings chains 10-12

Q4 2028 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q4 2028	4,641
2	Q1 2029	5,052
3	Q2 2029	5,242
4	Q3 2029	5,359
Sum	Forward four quarters	20,294
Valuation	Sum x 20 / 388 million shares	USD 1,046 per share

Q1 2029 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q1 2029	5,052
2	Q2 2029	5,242
3	Q3 2029	5,359
4	Q4 2029	5,286
Sum	Forward four quarters	20,939
Valuation	Sum x 20 / 388 million shares	USD 1,079 per share

Q2 2029 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q2 2029	5,242
2	Q3 2029	5,359
3	Q4 2029	5,286
4	Q1 2030	5,820
Sum	Forward four quarters	21,707
Valuation	Sum x 20 / 388 million shares	USD 1,119 per share

Explicit earnings chains 13-15

Q3 2029 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q3 2029	5,359
2	Q4 2029	5,286
3	Q1 2030	5,820
4	Q2 2030	6,068
Sum	Forward four quarters	22,533
Valuation	Sum x 20 / 388 million shares	USD 1,161 per share

Q4 2029 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q4 2029	5,286
2	Q1 2030	5,820
3	Q2 2030	6,068
4	Q3 2030	6,315
Sum	Forward four quarters	23,488
Valuation	Sum x 20 / 388 million shares	USD 1,211 per share

Q1 2030 valuation: all four earnings terms

Term	Operating quarter	Adjusted OP
1	Q1 2030	5,820
2	Q2 2030	6,068
3	Q3 2030	6,315
4	Q4 2030	6,368
Sum	Forward four quarters	24,570
Valuation	Sum x 20 / 388 million shares	USD 1,266 per share

Bounded diagnostics and their meaning

Diagnostic	Q4 2026 ceiling	Q4 2027 ceiling	Q4 2026 client new revenue	Q4 2026 client new OP
Earlier USD 30 endpoint; client off	556	801	0	0
Selected USD 40; client off	565	819	0	0
Nearline probability zero	545	780	3	2
Client effective-quantity index 10% lower	565	819	0	0
Client effective-quantity index 10% higher	566	819	7	5
Upper reported USD 70 endpoint	594	873	3	2

The earlier endpoint control is a comparison rather than the active case. The selected USD 40 case already includes the probability-weighted branch once. The USD 70 endpoint is an illustrative upper observation, without extra active weight. The quantity bounds alter the client overlap proxy, not cloud shipments,

manufacturing capacity or qualification. Client changes expire before Q4 2027, explaining the unchanged active target across those client diagnostics.

A higher diluted denominator would mechanically lower the ceiling without changing operating earnings. The unchanged denominator convention does not forecast that repurchases will eliminate every future award or convertible effect. Marginal cost rates are LOW-confidence allowances; neither source offers a new bill of materials or producer net-price sample to validate them.

Risks, failure channels and cross-company overlap

Risk class	Score	Failure channel
Technology / qualification	-1	HAMR reliability, yields or 40TB execution delay accepted shipments and raise cost.
Demand / customer concentration	-1	Large buyers defer deployment, digest inventory or negotiate away storage rents.
Competition / substitution	-1	Seagate or Toshiba density/supply rises; cheaper SSDs limit HDD applications or renewal pricing.
Financial / input cost / policy	-1	Higher capex, memory inputs, convert settlement, funding costs or restrictions reduce cash returns.

WDC and Seagate share the HDD demand cycle, but density, yield, customer acceptance and contract exposure must reconcile at company level. One supplier's market-share gain is not a simultaneous gain for all suppliers. Toshiba's storage-capacity metric is distinct from finished-drive units.

Sandisk, Kioxia, Samsung, SK hynix and Micron share AI and storage demand across different tiers. WDC receives no automatic NAND-price multiplier and consolidates no separated Sandisk earnings. Expensive NAND can support HDD relative economics while raising drive component costs. Optical and compute spending also shares the same customer funding and power constraints; adding all sector spending wallets to HDD sales would duplicate exposure.

Dated monitoring and source completeness

Checkpoint	Evidence that matters	Counter / response
October 22, 2026 results	Accepted EB, realized USD/TB, guide, client price versus unit mix and component costs	A dealer premium without producer realization reduces transmission; no unreported beat assumed today.
Q1 2027	40TB near half of nearline EB; first qualified HAMR deliveries	Reduce the affected conversion if acceptance misses.
Q2-Q4 2027	HAMR yields, accepted mix, UltraSMR overlap and renewed contract terms	Separate density from unit capacity and old from newly priced orders.
November 16, 2026 / next filing	Convertible settlement, cash and actual diluted shares	Revise the share convention only with the relevant settlement/award economics.
2028-30	Data retention, installed capacity, manufacturing expense and inventory	Revisit LOW-confidence extensions and fade if supply/demand or cost transmission changes.

Both newest completed management earnings discussions were read in full: April 30 and August 5, including prepared remarks, all Q&A and operator closing. The full September 9 Citi discussion, audience questions and closing were also read. Preserved text is a Quartr / StockAnalysis reproduction of management speech, rather than an issuer-issued transcript. Official audio was not replayed. The next-call announcement was read on the issuer site; its fresh raw download returned HTTP 403. These access/completeness distinctions remain explicit. [WDC-3; WDC-4; WDC-5; WDC-9]

The owner-supplied bank report has 28 preserved pages. Material pricing and timing pages were visually checked; narrative pages 1-14 were reviewed, with original pages 2-3 embedded here. Dense bank model matrices were not reconstructed and their forecasts are not imported. Unreported producer realization, non-nearline units/ASP, actual product margins, the exact cloud EB split and a full forward cash/asset schedule remain gaps. Near-term consolidated guidance has HIGH source confidence; product cost and longer-dated operating allocations remain LOW.

Source register

HDD-P2: Morgan Stanley Research, Erik W. Woodring and listed team, October 20, 2026 at 03:44 UTC. Owner-supplied 28-page earnings preview; original SHA-256 f1becebecbd71067332d91e407d2a382b1973e5f0d8c413dde427f30d770902b. Pages 2-3 provide the attributed pricing checks and hypothetical mix example. Original licensed exhibits are retained privately. This is a bank preview rather than company results.

WDC-1 | August 5, 2026. FY2026 Q4 release and GAAP reconciliation.

WDC-2 | August 14, 2026. FY2026 Form 10-K.

WDC-3 | August 5, 2026. FY2026 Q4 management call reproduction.

WDC-4 | April 30, 2026. FY2026 Q3 management call reproduction.

WDC-5 | September 9, 2026. Citi management conference reproduction.

WDC-6 | September 21, 2026 close. Historical quote snapshot retained from dated research.

WDC-7 | September 14, 2026. Convertible redemption notice.

WDC-8 | September 15, 2026. WD-sponsored AI data research.

WDC-9 | October 8, 2026. Upcoming October 22 results announcement.

TOS-1 | October 2, 2026. Toshiba first shipments from expanded nearline line.

D1 | July 30, 2026. Amazon Q2 management capex outlook.

D2 | July 29, 2026. Microsoft FY2026 Q4 management call.

D3 | July 22, 2026. Alphabet Q2 management capex outlook.

D4 | July 29, 2026. Meta Q2 financial results.

Inherited storage-capacity share exhibit: Nikkei / Techno Systems Research, 2025 capacity basis, preserved from the October 8 artifact. Its dated shares are contextual and supply no new WDC shipment or revenue amount. All source originals and unavailable/gated flags remain in the local source packet.

Technical precision supplement

Precision supplement - reconciliation only. Main tables and narrative use whole financial numbers. The following exact technical values preserve small pricing premiums, timing factors and four-term arithmetic; they are not additional forecasts.

Technical control / dated source	Exact value
Historical September 21 close; USD per share	448.170000
Historical August 7 basic shares M	360.540944
Cloud EB allocation proxy	207.900000
Cloud sales / quarter-basket coefficient	0.018212
2028 capex / revenue cross-check	0.065000
2030 capex / revenue cross-check	0.055000
November 16 redemption principal; USD M	109.505000
Active same-quarter ceiling before; USD	800.932350
Active same-quarter ceiling after; USD	819.032566
Active same-quarter change; percent	2.259893
Active F4 OP after; USD M	15,889.231773
New F4 OP vs before; USD M	351.144180

Exact pricing, phase and competitor reconciliation

Quarter	Reference USD/TB	Toshiba absolute loss USD/TB	Phase	Weighted scarcity USD/TB	Active USD/TB
Q3 2026	16.367713	0.000000	0.000000	0.000000	16.367713
Q4 2026	16.776906	0.000000	0.250000	0.174173	16.951079
Q1 2027	17.196328	0.000000	0.500000	0.342055	17.538384
Q2 2027	17.540255	0.000000	0.750000	0.505344	18.045599
Q3 2027	17.891060	0.000000	1.000000	0.663268	18.554328
Q4 2027	18.159426	0.045399	1.000000	0.655217	18.769245
Q1 2028	18.159426	0.090797	1.000000	0.655217	18.723846
Q2 2028	18.159426	0.136196	1.000000	0.655217	18.678448
Q3 2028	18.159426	0.181594	1.000000	0.655217	18.633049
Q4 2028	18.159426	0.181594	1.000000	0.655217	18.633049
Q1 2029	18.113856	0.090569	0.750000	0.492438	18.515725
Q2 2029	18.068401	0.090342	0.500000	0.328974	18.307033
Q3 2029	18.023059	0.090115	0.250000	0.164827	18.097771
Q4 2029	17.977832	0.089889	0.000000	0.000000	17.887943
Q1 2030	17.887260	0.089436	0.000000	0.000000	17.797824
Q2 2030	17.797145	0.088986	0.000000	0.000000	17.708160
Q3 2030	17.707484	0.088537	0.000000	0.000000	17.618947
Q4 2030	17.618275	0.088091	0.000000	0.000000	17.530184

Exact finite client-price residual

Quarter	Conditional R	Existing pool removed	Residual	Weighted new R
Q4 2026	517.500000	480.000000	37.500000	2.812500
Q1 2027	517.500000	475.000000	42.500000	3.187500
Q2 2027	517.500000	490.000000	27.500000	2.062500

Quarter	Weighted new R	New COGS	New opex	New OP
Q4 2026	2.812500	0.421875	0.084375	2.306250
Q1 2027	3.187500	0.478125	0.095625	2.613750
Q2 2027	2.062500	0.309375	0.061875	1.691250

WDC / OCTOBER 21, 2026 / FRAMEWORK 8.0

Exact four-term valuations 1-5

Valuation quarter	Four adjusted OP terms; USD M	F4 OP	Multiple / shares M	Ceiling USD
Q3 2026	1,880.500000 + 2,204.099338 + 2,554.241466 + 2,944.863313	9,583.704117	20 / 388	494.005367
Q4 2026	2,204.099338 + 2,554.241466 + 2,944.863313 + 3,269.719126	10,972.923242	20 / 388	565.614600
Q1 2027	2,554.241466 + 2,944.863313 + 3,269.719126 + 3,596.246728	12,365.070633	20 / 388	637.374775
Q2 2027	2,944.863313 + 3,269.719126 + 3,596.246728 + 3,812.747370	13,623.576536	20 / 388	702.246213
Q3 2027	3,269.719126 + 3,596.246728 + 3,812.747370 + 4,076.973131	14,755.686355	20 / 388	760.602389

Exact four-term valuations 6-10

Valuation quarter	Four adjusted OP terms; USD M	F4 OP	Multiple / shares M	Ceiling USD
Q4 2027	3,596.246728 + 3,812.747370 + 4,076.973131 + 4,403.264544	15,889.231773	20 / 388	819.032566
Q1 2028	3,812.747370 + 4,076.973131 + 4,403.264544 + 4,641.039896	16,934.024941	20 / 388	872.887884
Q2 2028	4,076.973131 + 4,403.264544 + 4,641.039896 + 5,051.767922	18,173.045494	20 / 388	936.754922
Q3 2028	4,403.264544 + 4,641.039896 + 5,051.767922 + 5,241.919447	19,337.991809	20 / 388	996.803702
Q4 2028	4,641.039896 + 5,051.767922 + 5,241.919447 + 5,359.461453	20,294.188718	20 / 388	1,046.092202

WDC / OCTOBER 21, 2026 / FRAMEWORK 8.0

Exact four-term valuations 11-15

Valuation quarter	Four adjusted OP terms; USD M	F4 OP	Multiple / shares M	Ceiling USD
Q1 2029	5,051.767922 + 5,241.919447 + 5,359.461453 + 5,285.995042	20,939.143863	20 / 388	1,079.337313
Q2 2029	5,241.919447 + 5,359.461453 + 5,285.995042 + 5,819.551952	21,706.927894	20 / 388	1,118.913809
Q3 2029	5,359.461453 + 5,285.995042 + 5,819.551952 + 6,067.517038	22,532.525485	20 / 388	1,161.470386
Q4 2029	5,285.995042 + 5,819.551952 + 6,067.517038 + 6,314.923202	23,487.987235	20 / 388	1,210.720991
Q1 2030	5,819.551952 + 6,067.517038 + 6,314.923202 + 6,367.709242	24,569.701435	20 / 388	1,266.479455